REPORT ON TURBOCHARGER FAILURE AT RAJOLA

Background:

The Turbocharger of MP2 of MPPL, Rajola got damaged on 12.12.2013. The cartridge assembly and stationary nozzle from the turbocharger of MP-1 was fitted in MP-2 and the engine was put back to operation on 14.12.2014. After running of 200 hrs, the engine was again tripped due to abnormal sound at 01:52 hrs. on dt. 28.12.2013 with black smoke. On opening, the turbocharger found to be damaged badly. Two turbochargers got damaged in quick succession in MP2. For detailed analysis & identification of the causes of failure and finding out the remedial measures for avoidance of such failures in future, management have advised undersigned to visit Rajola and analyze the case as single member fact finding committee vide order no.. dated 7.1.2013.

Undersigned visit Rajola from 9.1.13 to 10.1.2013, along with Mr. S.P.Saxena DGM, Sendra and inspected the damaged parts, reviewed records from log books and interacted with the maintenance team consisting of Sh. S.K.Verma, Sh.D.K.Verma, Foreman & Sh. Amrish Pandey Engg. Assistant and collected the following facts:

DETAILS OF INCIDENCE:

On 28.12.2013, MP-2, Rajola was tripped due to abnormal sound at 1:52 hrs. with black smoke. At the time of tripping engine was running at 730 RPM with calculated load of 98%. Exhaust temperature:350, 365, 355, 374, 349, 357: T/C entry/exit – 445, 445 & 210. Charge air pressure of the engine was 2.18 Kg/cm2. CRH of engine on dt. 28.12.2013 was 75910. After major overhaul of T/C, it rendered service of only 203 hrs.

The Turbocharger of MP2 was damaged in the similar fashion on dt. 12.12.2013.

The maintenance activities carried out while fitting the cartridge assembly and stationary nozzle of MP-1 in MP-2 was reviewed and found as follows:-

- All cylinder heads dismantled and inlet & exhaust valve and valve seat inspected. Everything was found intact.
- Water leakage was observed in two cylinder head from Injector sleeve and it was replaced by refurbished cylinder heads.
- All exhaust piping checked & inspected and everything was found normal.
- Thermocouple was checked and found O.K.
- Engine was put back to operation on 17.12.2013 at CRH 75707.

On 28.12.2013, the engine again failed due to failure of the turbocharger after operation of 203 hrs. The engine was dismantled on 1.1.2014

DETAILS OF TURBOCHARGER

Make: Napier

Model: 295

Date of commissioning: 1.10.1997

Total run hours of turbocharger 1, till failure: 67820 (failed on 12/12/2013)

Total run hours of turbocharger 2, till failure: 77695 (failed on 28/12/2013)

DETAILS OF DAMAGE OBSERVED IN THE TURBOCHARGER:

After dismantling of the turbocharger, following damages were found in the rotor assembly:

- One blade of T/C turbine was blown off. Remnants of the blade were also not found.
- A scratch mark on the periphery of stationary nozzle towards turbine entry was found.
- Turbine seal end bush got cracked (snaps attached)
- Impeller (Compressor wheel) got damaged (snap attached)
- Bush bearings and seal rings of both side badly affected.
- Rubbing marks of the face of thrust collar also appeared.

One tooth of turbine blades damaged

Scratch mark appeared on the periphery

Damaged impeller

Crack appeared on the this position of turbine seal end

After opening the stationary nozzle from double entry, it was found that other side (towards exhaust pipe) of the stationary ring was in perfectly good condition.

INSPECTION CARRIED OUT ON MP2

In order to avoid presence of any loose parts in engine following inspection/ maintenance activities are carried out before fitting of new rotor assembly.

1. All the exhaust pipes dismantled, checked and found O.K. It has been thoroughly cleaned
2. Thermocouple checked and found O.K.
3. Bellow checked and found O.K.
4. Cylinder head were dismantled after just 203 hrs. and inspected properly.
5. Oil bath filter checked and all the neoprene flappers found intact.
6. Fuel pump checked & found ok.
7. Injectors tested at 4000PSI & all injector nozzles are replaced..
8. Tappet of the engine reset
9. Bend before charger cooler was dismantled and checked for any foreign particles. Broken pieces of impeller of turbo charger were found at the front face of charge air cooler.

From the nature of damages it is observed that chances of entry of any foreign particle from engine side are very remote. Had it been so then there should have been damage mark on the nozzle housing in the turbine inlet casing side. Hence damage to turbine bladed might have been from failure of some internal parts of the rotor assembly itself

COMPARISON OF THE FAILURES OF BOTH THE TURBOCHARGERS:

The nature of failures observed at both the failed turbochargers at MP2 found identical as follows:

Turbocharger failed on 12.12.2013 on MP2	Turbocharger failed on 28.12.2013 on MP2

• 3 blades of the turbine found damaged (1 from the root and other at the tips • The lancing wire found broken at the 3 blades • Stationary nozzle towards the turbine side found damaged • No damage mark on the nozzle housing in the turbine inlet casing side. • Impeller wheel (compressor wheel) found damaged (rubbing on the impeller with labyrinth seal plate & compressor outlet casing) • Turbine end seal bush found damaged (collar broken)	• 1 blade of the turbine found damaged (from the root) • The lancing wire found broken at the blade • Small scratch was found on the stationary nozzle towards the turbine side found damaged • No damage mark on the nozzle housing in the turbine inlet casing side. • Impeller wheel (compressor wheel) found damaged (rubbing on the impeller with labyrinth seal plate & compressor outlet casing) • Turbine end seal bush found with crack mark (collar cracked)

The identical failure of the rotor assembly components suggests common sources of problem.

Possible sources of problem:

1. Problem from engines
    a. Ingress of failed engine components into turbocharger(inlet/exhaust valve, valve seats, cylinder head component, injector nozzle etc)
    b. Extreme operating condition (Overloading of engine, High exhaust temperature etc.)
2. Fitment problem
    a. Improper procedure
    b. Improper clearance

3. Problem with turbocharger itself
    a. Inferior components
    b. Fatigue of components

Analysis:

1. From inspection of the engine components after dismantling of the engine, no abnormalities observed after failure of both the turbochargers and even no debris could be found from the exhaust manifold. The inlet /exhaust valve, valve seats, cylinder head component, injector nozzle etc are checked thoroughly and all parts found intact. Hence chances of entry of any engine components inside the turbocharger found remote.

    The operating parameters of all the three engines MP1, MP2 & MP3 at Rajola have been reviewed and found as follows:

		MP1	MP2	MP3

		Before/ After cooler	Before/ After cooler	Before/ After cooler
Charge air Temp. in Degree C	Dec/2013	138/50	137/49	132/27
Charge air Pr (Kg/cm2)		1.7	2.15 (Slightly on higher side)	1.7
Load		92% (impeller trimmed in March/2013)	98%	94%
Exhaust temperature (Maxm) (in Degree C)		383	374	429
Temp at TC turbine end (in Degree C		461/452	445/445	506/506
Temp at TC exit		302	270	353

Hence operating condition of all the three engines found almost identical and failure of turbocharger in MP2 cannot be attributed to extreme operating condition.

2. Fitment Issues-Improper procedure:

   a. All procedure for fitment of turbochargers has been followed as per OEM manual. All schedule maintenances carried out as per OEM manual and details of work carried out are checked from maintenance history records and no deviation found. As per OEM, the complete rotor assembly to be replaced by a new one after completion of 48000 hrs.(Ref. Maintenance & Inspection manual Napier 295 page 33) which has not been followed.

   b. Clearances: The maintenance record shows the practice of taking the required clearances are existing at station and followed. For Turbocharger fitted in MP1 following clearances have been maintained.

	Range as per manual	Actual reading of TC as on 7.1.2014
Turbine wheel/ cone clearance	3.25 to 3.90 mm	3.60 mm
Impeller axial clearance	0.61 to 0.81 mm	0.67 mm
Impeller/Labyrinth clearance	0.28 to 0.33 mm	0.33 mm
Thrust clearance	0.16 to 0.24 mm	0.14 mm

Now for repair of MP2 turbocharger of MP2 (failed on 28.12.13), the cartridge assembly was taken out from MP1 and fitted in MP2 as it is and hence no clearances were taken.

However the new turbocharger fitted in MP2 on 8.1.2014 has been checked for clearances and reading recorded in the register are as follows:

	Range as per manual	Actual reading of TC as on 7.1.2014
Turbine wheel/ cone clearance	3.25 to 3.90 mm	3.90 mm
Impeller axial clearance	0.61to 0.81 mm	0.72 mm
Impeller/Labyrinth clearance	0.28 to 0.33 mm	0.29 mm
Thrust clearance	0.16 to 0.24 mm	0.18 mm

It is observed that the practices for taking all required clearances are existing and being followed properly.

3. Problem with turbocharger:

   All turbocharger spares are procured from OEM and no indigenous supply has been made. With the components supplied by OEM, the quality issues of the material can be ensured.

   The turbocharger failed in MP2 on 28.12.2013 has completed service of 77695 hrs and the turbocharger failed on same MP2 on 12.12.2013 has completed 67820 hours.

   As per OEM, the service life of the turbine rotor assembly (cartridge) is 48000 hours. The rotor assembly consists of turbine blades with lancing wire, rotor shaft, thrust collar, impeller, washers etc. The turbine blade & shaft assembly is an integral part and cannot be repaired and re-assembled. The turbine blades also once damaged cannot be repaired.

   For rotor assembly following checks are suggested in OEM
   - Cleaning
   - Visual inspection for damage
   - Dynamic balancing at every 32000 hrs or whenever any major parts are replaced

<u>Conclusion:</u>

At Rajola, the rotor assembly is inspected by dynamic balancing at every 9000 hours and so far the turbocharger has performed well despite running beyond 48000 hrs. But with the use the rotor blades might have been fatigued and failed. The blades which are subjected to maximum stress have failed on fatigue and sheared off from the root and further damaged the other parts.

<u>Recommendations:</u>

1. There is no remaining life assessment measures exists to ascertain the life of the rotor assembly
2. It is recommended that some non destructing testing may be carried out for assessing the health of the rotor assembly after expire of recommended service life of 48000 hours before re-use or
3. The rotor assembly be replaced at every 48000 hrs as recommended by OEM
4. Speed probe to be installed at the turbocharger as recommended by OEM for measuring the actual speed of the rotor assembly.

NOTE SHEET

WRPL/RAJ/MECH/MP-2/BDR 11.01.2014

SUB : Report of damage of Turbocharger of MP-2 on dt. 28.12.2013

Preliminary Observations :

MP-2, Rajola was tripped due to abnormal sound at 01:52 hrs. on dt. 28.12.2013 with black smoke. At the time of tripping engine was running at 730 RPM with calculated load of 98%. Exhaust temperature – 350, 365, 355, 374, 349, 357: T/C entry/exit – 445, 445 & 210. Charge air pressure of the engine was 2.18 Kg/cm2. CRH of engine on dt. 28.12.2013 was 75910. After major overhaul of T/C, it rendered service of only 203 hrs.

Previous History :

The Turbocharger of same engine was damaged in the similar fashion on dt. 12.12.2013. The cartridge assembly and stationary nozzle of MP-1 was fitted in MP-2 after following inspection.

1. All cylinder heads dismantled and inlet & exhaust valve and valve seat inspected. Everything was found intact. Albeit, water leakage was observed in two cylinder head from Injector sleeve and it were replaced by refurbished cylinder heads.
2. All exhaust piping checked & inspected and everything was found normal.
3. Thermocouple was checked and found O.K.

Engine was kept in operation on dt. 17.12.2013 at CRH 75707.

DAMAGES OCCURRED :

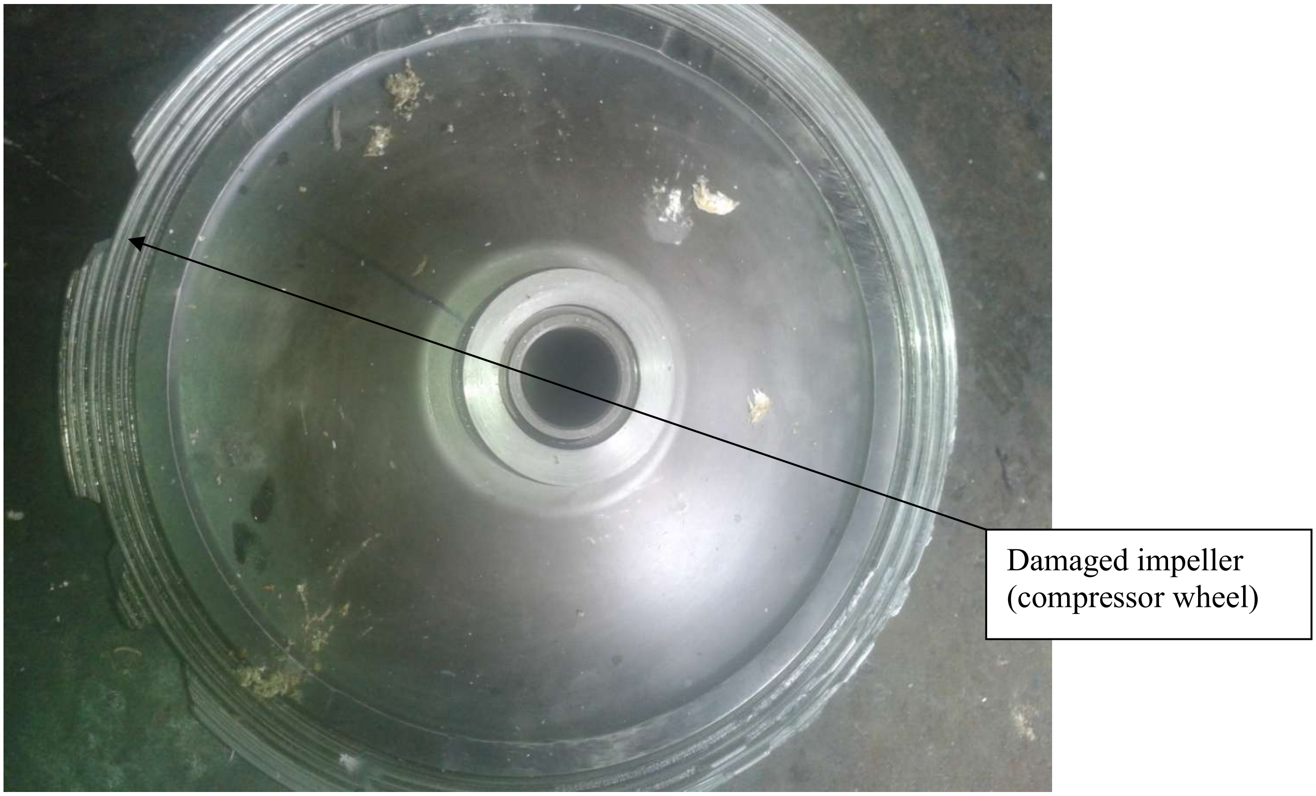
After dismantling following damages were found :

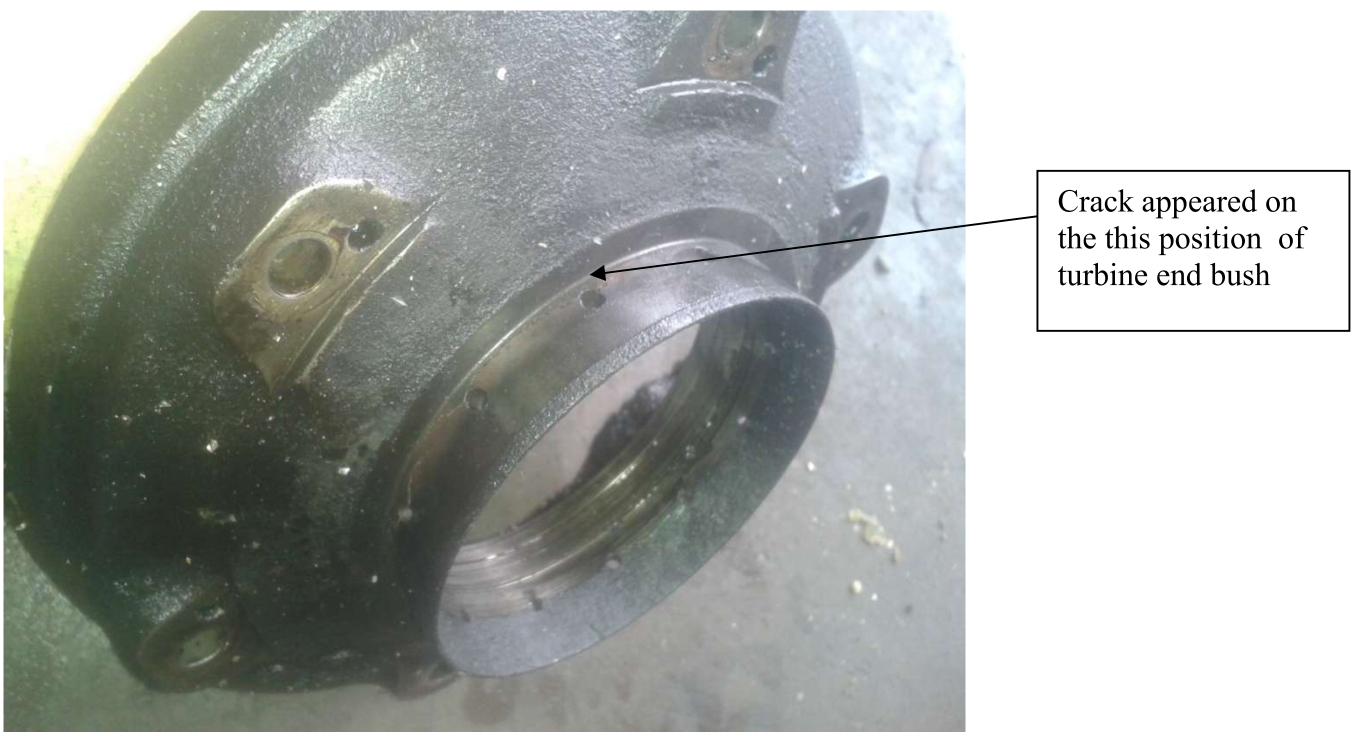
1. Prima facie, one blade of T/C turbine was blown off. Remnants of the blade was also not found.
2. A scratch mark on the periphery of stationary nozzle towards turbine entry was found.
3. Turbine seal end bush got cracked (snaps attached)
4. Impeller (Compressor wheel) got damaged (snap attached)
5. Bush bearings and seal rings of both side badly affected.
6. Rubbing marks of the face of thrust collar also appeared.

One teethof turbine blades damaged along with lancing wire

Scratch mark appeared on the periphery of nozzle assembly.

After opening the stationary nozzle from double entry, it was found that other side (towards exhaust pipe) of the stationary ring was in perfectly good condition.


Damaged impeller
(compressor wheel)


Crack appeared on
the this position of
turbine end bush

Action Taken:

As per advice, following inspection/ maintenance activities are being carried out before fitting of new rotor assembly.

1. All the exhaust pipes dismantled, checked and found O.K. It has been thoroughly cleaned also.
2. Thermocouple checked and found O.K.
3. Bellow checked and found O.K.
4. Cylinder head were dismantled just 203 hrs. and inspected properly. So it was not checked.
5. Oil bath filter checked and all the neoprene flappers found intact.
6. Fuel pump checked.
7. Injectors re-calibrated and all the nozzles replaced.

Parts replaced of Turbocharger :

1. **New Rotor assembly along with thrust Collar**
   No. depicted as follows
   [Turbine rotor shaft (New)
   BJ 4026/69/416796
   BJ 4126/07/5/909612 Aug.'13
   Comp. wheel (New) Dec.'13]
2. **Nozzle assembly**
3. **Bearing of both side**
4. **Seal ring of both side**
5. **Seal ring bush of both side**
6. **Labyrinth seal plate and its peelable shim**

Clearances are maintained as follows :

1. **Turbine wheel / Cone clearance – 3.9 mm**
2. **Impeller total float – 1.01 mm**
3. **Impeller axial clearance – 0.72 mm**
4. **Impeller/ Labyrinth clearance – 0.29 mm**
5. **Compressor end bearing thrust clearance – 0.18 mm**

NOTE : *Representative from M/s Napier also visited at site on dt. 31.12.2013 and 01.04.2014. His report attached.*

Engine was started on 10.01.2014 after no load running of 01 hrs.

(S. K. Verma)
DMNM, WRPL, Rajola

COM, WRPL, Rajola